

Research Profile

Researcher in systems and control with training in automation and applied mathematics. My research focuses on model predictive control, distributed control, and learning-based methods for constrained and networked dynamical systems.

Research Appointments

- Research Assistant, Robotics and Autonomous Systems** Feb. 2026 – Present

The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China

Supervisor: Prof. Yiding Ji. Research on distributed predictive control, multi-agent systems, and learning-assisted control.
- Research Assistant, Elmore Family School of Electrical and Computer Engineering** Oct. 2024 – Sep. 2025

Purdue University, West Lafayette, IN, USA

Supervisor: Prof. Philip E. Paré. Research on physics-informed learning and model predictive control.

Education

- B.Eng. in Automation and B.S. in Mathematics and Applied Mathematics** Sep. 2021 – Jun. 2025

South China University of Technology, Guangzhou, China

Supervisor: Prof. Langwen Zhang. GPA: 3.75/4.00; rank: 3/21. Outstanding Undergraduate Thesis, ranked 1/335.

Research Interests

- Model predictive control** Robust and event-triggered MPC; recursive feasibility; safety and stability of constrained systems.
- Distributed control** Leader-following coordination; heterogeneous multi-agent systems; switching topologies and communication attacks.
- Learning-assisted control** Physics-informed learning, reinforcement learning, online state/parameter estimation, and data-driven modeling for control.

Publications

- J1** A. Zhong, W. Lu, L. Zhang, and Z. Bao, “Adaptive event-triggered MPC for linear parameter-varying systems with state delays, actuator saturation and disturbances.”

Journal of Process Control, vol. 167, 103865, 2026. doi:10.1016/j.jprocont.2026.103865
- J2** A. Zhong, B. She, and P. E. Paré, “A physics-informed neural networks-based model predictive control framework for SIR epidemics.”

IEEE Open Journal of Control Systems, vol. 5, 2026. doi:10.1109/OJCSYS.2025.3641369
- J3** A. Zhong, W. Lu, and L. Zhang, “Resilient distributed model predictive control of state-delayed linear parameter varying systems with quantitative communication against denial of service attacks.”

Asian Journal of Control, vol. 27, no. 2, pp. 728–740, 2025. doi:10.1002/asjc.3465
- C1** A. Zhong, S. Zhou, and Y. Ji, “Distributed leader-following model predictive control for constrained heterogeneous multi-agent systems.”

Accepted by the *IEEE Conference on Decision and Control (CDC)*, 2026.

Patents

- P1** A. Zhong et al., “Learning-based predictive control method for autonomous vehicles using long short-term memory networks.”

Chinese invention patent ZL 2024 1 0055698.0, granted, 2024.
- P2** A. Zhong et al., “Off-policy output feedback data-driven Q-learning control method based on VI strategy.”

Chinese invention patent, under substantive examination, 2024.

Honors and Service

- Outstanding Undergraduate Thesis, South China University of Technology, 2025 (ranked 1/335).
- Haopeng Technology Diligence Scholarship, 2024 (CNY 10,000; ranked 1/30).
- Meritorious Winner, Interdisciplinary Contest in Modeling, 2022; team leader; top 6.44%.
- Sunwoda New Energy Technology Innovation Scholarship, 2022; Guangdong Undergraduate Mathematics Competition, Provincial Second Prize, 2021.
- Reviewer: IEEE Conference on Decision and Control and IEEE Transactions on Industrial Cyber-Physical Systems, 2026.

Technical Skills

Programming	Python, MATLAB, C/C++, LaTeX.
Optimization and ML	YALMIP, MOSEK, CVX, PyTorch, TensorFlow; quadratic programming, LMIs, robust optimization.
Languages	Mandarin (native); English (TOEFL MyBest 99: R28/L24/S21/W26).